**4** **(a)** State the *principle of superposition*.

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.....

..... [1]

**(b)** Noise-cancelling headphones use both Active Noise Control and passive sound-proofing to reduce undesired ambient sounds reaching the ears of the user. In Active Noise Control, a microphone detects ambient noise and a noise-cancellation speaker emits a corresponding "anti-noise" that undergoes destructive interference with the noise.

Suggest the features that the "anti-noise" should have in order to achieve the optimal noise-cancelling effect.

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..... [2]

**(c)** A student measured the waveforms of the noise and the anti-noise. The following observations were made:

1. There is a 1.0 ms time-lag between the noise and the anti-noise.
2. The amplitudes of the noise and anti-noise were effectively similar.

A sample noise waveform is shown in Fig. 4.1.

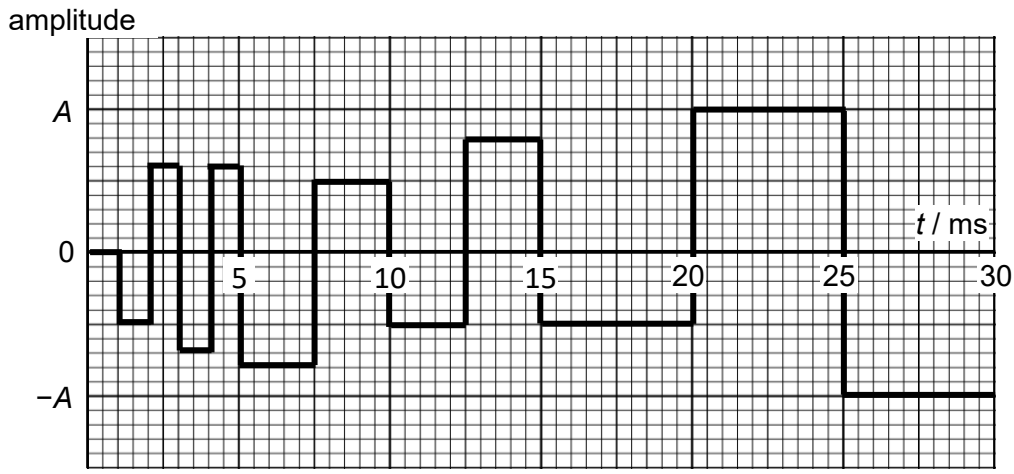


**Fig. 4.1**

**(i)** On Fig. 4.2, sketch the anti-noise waveform for 15 ms < $t$ < 30 ms. [2]

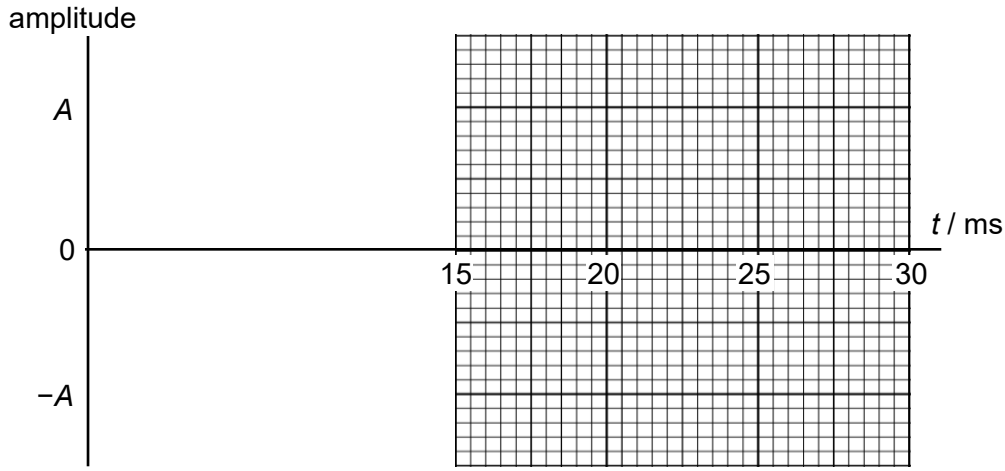


**Fig. 4.2**

**(ii)** The final resultant waveform reaching the ears is shown in Fig. 4.3.

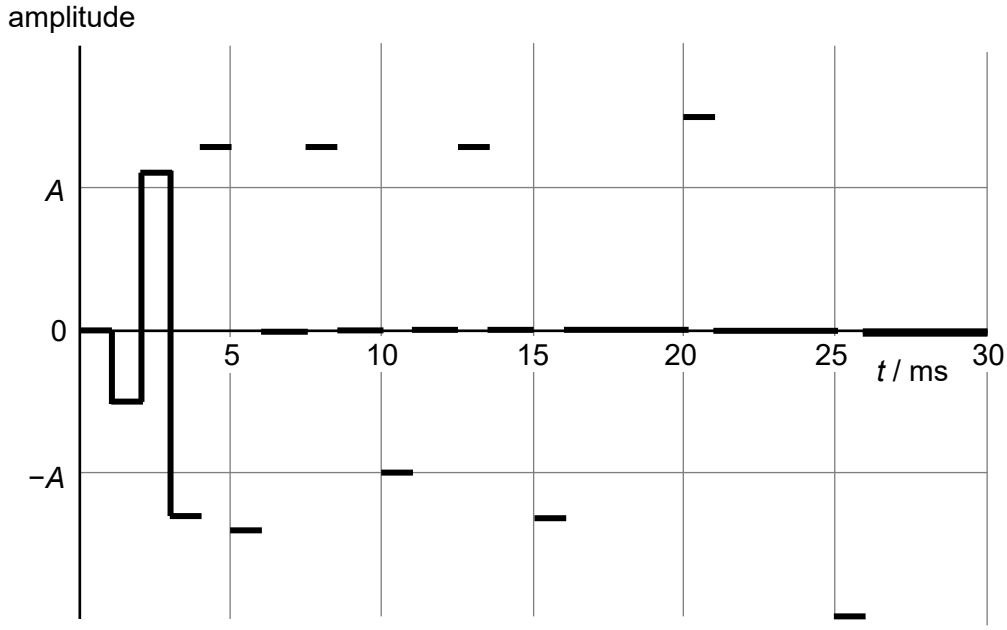


**Fig. 4.3**

Using Fig. 4.3, suggest if Active Noise Control is better suited for noises of higher or lower frequencies.

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..... [2]